

DHAMODHARAN S

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PROFESSIONAL SUMMARY

AI Engineer specialising in RAG Systems, LLM fine-tuning (LoRA/QLoRA), Agentic AI and Multi-agent systems. Built production-grade pipelines reducing analyst effort by **90%** and VRAM usage by **70%** on free hardware. Actively exploring research in LLM cognitive capabilities (reasoning, memory, and planning). Proficient in Python, PyTorch, LangChain, and LangGraph with hands-on MLOps exposure via Docker, Git, and AWS.

EXPERIENCE

Infosys Springboard

Nov 2025 – Jan 2026

AI Intern

- Developed **PlantDocBot**, an end-to-end AI system for plant disease diagnosis combining computer vision and NLP-based symptom analysis.
- Trained CNN models (ResNet, MobileNet) on PlantVillage & PlantDoc datasets achieving **15%** improvement in classification accuracy; fine-tuned BERT for symptom-based disease prediction and deployed via Streamlit.

EDUCATION

Sri Krishna College of Technology, Coimbatore, Tamil Nadu

2023 – 2027

B.E., Artificial Intelligence and Machine Learning

GPA: **8.31** / **10.0** (up to **5th** Semester)

TECHNICAL SKILLS

Core AI / ML: LLM Fine-Tuning, LoRA, QLoRA, vLLMs, Knowledge Distillation, Agentic AI, Multi-Agent Systems, RLHF, GRPO, RAG Systems, Vectorless RAG, Computer Vision, Deep Learning, Machine Learning

Frameworks & Tools: PyTorch, TensorFlow, LangChain, LangGraph, LangSmith, Unsloth, HuggingFace, PageIndex, CrewAI, spaCy

Vector & Graph Databases: ChromaDB, Pinecone, Weaviate, FAISS, Neo4j, MySQL, MongoDB

MLOps & Cloud: Docker, Git, MLflow, CI/CD, AWS (EC2, S3, SageMaker, Bedrock), Azure ML

Web, APIs & Automation: FastAPI, Flask, React, JavaScript, REST APIs, Groq API, Gemini API, n8n, Apache Airflow, Make

Programming: DSA and Algorithms, Python, Java, C++

PROJECTS

1. PaperCompass — Vectorless Multi-Paper RAG with Agentic Retrieval

[GitHub](#)

LangGraph, Groq, NVIDIA NIM, PyMuPDF, Pydantic, Chainlit

- Designed a vectorless RAG pipeline parsing PDFs into hierarchical section trees via an **8-step** layout-aware pipeline, replacing embeddings with LLM-driven section indexing to solve chunking's loss of document structure and context.
- Orchestrated a **4-node** LangGraph agent (router, tool executor, synthesiser, citation builder) across **5** papers per session with **3-hop** contradiction detection, delivering fully cited, page-referenced answers via Chainlit.

2. LLM Metacognitive Calibration Study

[GitHub](#)

Python, PyTorch, Transformers, scikit-learn, Sentence-Transformers

- Built a **170-item** benchmark (TriviaQA, TruthfulQA, SQuAD v2) to test if Qwen2.5 (**1.5B/7B**) models know when they're wrong, comparing **4** confidence signals: verbalized self-report, token entropy, hidden-state probing, self-consistency.
- Trained a CV-regularized logistic probe on pre-generation hidden states, achieving **AUROC 0.79** and reducing overconfidence-on-mistakes from **100%** (verbalized) to **19%** — proving internal states encode truer confidence than stated confidence.

3. FableDistill — Reasoning Distillation: Fable-5 Traces into Qwen3-4B

[GitHub](#)

Qwen3-4B, QLoRA, Unsloth, SFT, Knowledge Distillation, Kaggle T4 GPU

- Distilled Claude Fable-5's chain-of-thought reasoning into Qwen3-4B via QLoRA SFT on **4,659** agentic sessions with cleartext `<think>` traces, training only **0.80%** of parameters ($r=16$, $\alpha=32$) on a free Kaggle T4 GPU using Unsloth with masking, addressing small LLMs' lack of planning and self-correction.
- Achieved **55%** training loss reduction ($1.957 \rightarrow 0.878$) over **583** steps in **1** epoch; resulting model branches into multiple hypotheses and self-corrects mid-trace within **~3 GB** VRAM.

4. Product Launch War Room: Multi-Agent Decision System

[GitHub](#)

Python, LangGraph, LangChain, Pydantic, LangSmith

- Orchestrated **4** specialised agents via LangGraph with **7** tools (z-score anomaly detection, sentiment clustering, composite risk scoring) across **9** metrics and **35+** feedback entries to solve teams' lack of real-time, data-driven launch frameworks.
- Delivered Proceed/Pause/Roll-Back decisions in **17.84s** avg latency with LangSmith tracing, cutting analyst effort by **~90%** at **~\$0.02** per run.

CERTIFICATIONS AND ACHIEVEMENTS

IBM Machine Learning with Python — Coursera

Topper + Elite Silver — Deep Learning, IIT Ropar (NPTEL)

5-Day AI Agentic Intensive Course — Kaggle and Google Cloud